Corresponding Parts of Congruent Triangles

Answer Key

High School Geometry

Quick Answers

1. 6
2. 15
3. (a) y = 8, (b) MN = 28 ft
4. (a) x = 17, (b) angle Z = 58
5. (a) n = 6, (b) perimeter = 49 m
6. (a) x = 15, (b) angle G = 57, —
7. (a) x = 8, (b) y = 7, (c) DE = 25 cm
8. (a) w = 7, (b) k = 10, —

What is verified

6 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry
HSG-CO.B – *problems 1, 2, 3, 4, 5, 6, 7, 8*
HSG-CO.C – *problems 6, 8*
Difficulty 1–4 of 5 across 17 tagged checks.

Common wrong answers

4. *If they got 74: paired Z with Y instead of C and copied the 74 degree angle*
5. *If they got 32: added only the two sides whose lengths were printed as numbers*

1. A machine shop cuts two congruent steel gusset plates, so that $\triangle ABC \cong \triangle DEF$. The shop drawing lists $AB = (2x+3)$ cm and $DE = 15$ cm. Find x.
Solution. The statement pairs A with D and B with E, so $\overline{AB}$ corresponds to $\overline{DE}$ and the two lengths are equal: $2x+3 = 15$. Subtracting 3 gives $2x = 12$, so $\boxed{x = 6}$

2. The two spars of a kite frame form congruent triangles with $\triangle PQR \cong \triangle STU$. The pattern lists $m\angle P = (5x-10)^\circ$ and $m\angle S = 65^\circ$. Find x.
Solution. P is paired with S, so $\angle P$ and $\angle S$ are corresponding angles and have equal measures: $5x - 10 = 65$. Adding 10 gives $5x = 75$, so $\boxed{x = 15}$

3. A footbridge is stiffened by two congruent gusset triangles with $\triangle JKL \cong \triangle MNP$. The fabrication drawing gives $JK = (3y + 4)$ ft and $MN = (5y - 12)$ ft. (a) Find y. (b) Find the length of $\overline{MN}$, in feet.
Solution (a). J pairs with M and K pairs with N, so $\overline{JK}$ corresponds to $\overline{MN}$:

$$3y + 4 = 5y - 12.$$

Subtract $3y$ from both sides: $4 = 2y - 12$. Add 12: $16 = 2y$, so $\boxed{y = 8}$
Solution (b). Substitute into the expression for MN: $MN = 5(8) - 12 = 40 - 12$, so $\boxed{MN = 28 \text{ ft}}$

Check: $JK = 3(8) + 4 = 28$ ft as well, as it must be.

---

4. A roof frame is built from two congruent trusses with $\triangle ABC \cong \triangle XYZ$. The plan gives $m\angle A = 48°$, $m\angle B = (4x + 6)°$, and $m\angle Y = 74°$. (a) Find x. (b) Find $m\angle Z$, in degrees.
Solution (a). B pairs with Y, so $4x + 6 = 74$. Subtract 6: $4x = 68$, so $\boxed{x = 17}$
Solution (b). Z pairs with C, so $m\angle Z = m\angle C$. The angles of $\triangle ABC$ sum to 180°, and $m\angle B = m\angle Y = 74°$, so

$$m\angle C = 180 - 48 - 74.$$

Therefore $\boxed{m\angle Z = 58°}$

---

5. A sailmaker cuts two congruent triangular panels with $\triangle RST \cong \triangle UVW$. The cutting list gives $RS = 12$ m, $ST = (2n + 5)$ m, $RT = 20$ m, and $VW = 17$ m. (a) Find n. (b) Find the perimeter of $\triangle UVW$, in metres.
Solution (a). S pairs with V and T pairs with W, so $\overline{ST}$ corresponds to $\overline{VW}$: $2n + 5 = 17$. Subtract 5: $2n = 12$, so $\boxed{n = 6}$
Solution (b). Congruent triangles have equal perimeters, so add the three sides of $\triangle RST$: $RS = 12$, $ST = 17$, $RT = 20$.

$$12 + 17 + 20 = 49.$$

Therefore $\boxed{P = 49 \text{ m}}$

---

6. A bicycle frame joint is reinforced with two congruent plates, $\triangle GHJ \cong \triangle KLM$, with $m\angle G = (3x + 12)°$ and $m\angle K = (5x - 18)°$. (a) Find x. (b) Find $m\angle G$, in degrees. (c) Explain how you know that $m\angle J = m\angle M$ without measuring either angle.
Solution (a). G pairs with K, so

$$3x + 12 = 5x - 18.$$

Subtract $3x$: $12 = 2x - 18$. Add 18: $30 = 2x$, so $\boxed{x = 15}$
Solution (b). $m\angle G = 3(15) + 12 = 45 + 12$, so $\boxed{m\angle G = 57°}$
Solution (c) — instructor-judged. A complete response says that the congruence statement pairs vertex J with vertex M, so $\angle J$ and $\angle M$ are corresponding angles of congruent triangles; by CPCTC corresponding parts of congruent triangles are congruent, so the two measures are equal and neither has to be measured. Award full credit only when both the correspondence and the CPCTC justification appear.

---

7. Two congruent triangular tent panels satisfy $\triangle ABC \cong \triangle DEF$. The pattern lists $AB = (4x - 7)$ cm, $DE = (2x + 9)$ cm, $BC = (3y + 1)$ cm, and $EF = (y + 15)$ cm. (a) Find x. (b) Find y. (c) Find the length of $\overline{DE}$, in centimetres.

Solution (a). $\overline{AB}$ corresponds to $\overline{DE}$:

$$4x - 7 = 2x + 9.$$

Subtract $2x$: $2x - 7 = 9$. Add 7: $2x = 16$, so $\boxed{x = 8}$

Solution (b). $\overline{BC}$ corresponds to $\overline{EF}$:

$$3y + 1 = y + 15.$$

Subtract y: $2y + 1 = 15$. Subtract 1: $2y = 14$, so $\boxed{y = 7}$

Solution (c). $DE = 2(8) + 9 = 16 + 9$, so $\boxed{DE = 25 \text{ cm}}$

Check: $AB = 4(8) - 7 = 25$ cm, matching DE as required.

---

8. A crane rests on two congruent support triangles with $\triangle PQR \cong \triangle XYZ$. The specification gives $PQ = (2w + 5)$ m, $XY = 19$ m, $m\angle Q = (7k - 4)°$, and $m\angle Y = 66°$. (a) Find w. (b) Find k. (c) The engineer claims that knowing only $PQ = XY$ and $m\angle Q = m\angle Y$ would already guarantee the two triangles are congruent. Explain why that is not enough, and name one additional measurement that would settle it.

Solution (a). $\overline{PQ}$ corresponds to $\overline{XY}$, so $2w + 5 = 19$. Subtract 5: $2w = 14$, so $\boxed{w = 7}$

Solution (b). $\angle Q$ corresponds to $\angle Y$, so $7k - 4 = 66$. Add 4: $7k = 70$, so $\boxed{k = 10}$

Solution (c) — instructor-judged. A complete response says that one pair of congruent sides together with one pair of congruent angles is not a triangle congruence criterion: infinitely many differently shaped triangles share one side length and one angle measure. The response must then name a sufficient addition — a second pair of congruent angles (giving ASA or AAS), or the second pair of sides that includes the given angle (giving SAS). Award full credit only when a valid criterion is named.